



MD RAFIUL ISLAM

AI ENGINEER

Computer Vision | Edge AI | Agentic Systems

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PROFESSIONAL SUMMARY

AI Engineer with 2+ years across production computer vision, edge deployment and applied AI research. Architected a multi-module surveillance architecture for BRAC Bank's 1,000-camera rollout and delivered real-time AI solutions for six enterprise clients across banking, logistics, manufacturing and retail. Hands-on with Python, PyTorch, YOLO, OpenCV, RTSP/GStreamer, NVIDIA Jetson, ONNX/TensorRT, FastAPI, Docker, LangGraph and hybrid RAG. Builds the full path from model and camera integration to temporal validation, APIs, evidence-backed alerts, Linux deployment, customer PoCs and technical pre-sales.

VALUE AT A GLANCE

- **2+ years** across applied AI engineering and research
- **1,000-camera** enterprise AI rollout
- **6 enterprise clients** across banking, logistics, manufacturing and retail
- **3 first-author papers** including a peer-reviewed Journal of Voice article
- **End-to-end ownership** from PoC and architecture to field deployment

EDUCATION

B.Sc. in Computer Science & Engineering

Daffodil International University | 2020-2024
CGPA 3.52 / 4.00

Thesis: Bengali voice-based psychological stability classification using spectrograms and hybrid CNN-ViT models.

HSC - BAF Shaheen College Dhaka
GPA 4.45 / 5.00 | 2018

SSC - Tamirul Millat Kamil Madrasha
GPA 5.00 / 5.00 | 2016

CORE EXPERTISE

Production CV - YOLO v8/v11/YOLOE, OBB, detection, segmentation, pose, OpenCV, ByteTrack, BoT-SORT, DeepSORT

OCR & Identity - PaddleOCR, RapidOCR, Bangla ANPR, ISO 6346 validation, InsightFace, FAISS

Edge AI - Jetson Orin Nano, ONNX, TensorRT, FP16/INT8, NVDEC, GStreamer

Video Systems - Multi-stream RTSP, CCTV/IP cameras, shared-memory frame bus, multi-GPU scheduling

Agentic AI - LangGraph, BM25/vector RAG, RRF, embeddings, tool calling, Ollama

Backend & MLOps - FastAPI, REST, Docker, systemd, SQLite/WAL, Git/GitLab, W&B, TensorBoard

IoT & Systems - Linux, multiprocessing, ESP32/UART, queues and restart-safe services

PROGRAMMING

• Python | SQL | Bash | Java

PROFESSIONAL EXPERIENCE

Machine Learning Engineer

Bondstein Technologies Ltd.

Enterprise computer vision, edge AI and customer solution engineering

Feb 2025 - Present
Dhaka

- **Architected a multi-module AI surveillance system for BRAC Bank** covering face recognition, lying detection, suspicious objects, unauthorized/restricted entry, camera tampering, behavioural monitoring and abusive-audio analysis across a 1,000-camera rollout.
- **Developed a shared-memory video frame bus** for the current 383-camera unified registry snapshot, decoding each RTSP stream once and supporting NVDEC/software fallback, camera health monitoring and multi-GPU workload planning.
- **Engineered validation-first false-alarm controls** using tracking, temporal voters, ROI and homography rules, pose geometry, identity logic, semantic verification and auditable evidence for high-consequence alerts.
- **Delivered edge and industrial AI solutions for five additional enterprise clients** spanning Jetson container OCR, industrial safety, leaf-buying SOP compliance, showroom intelligence and product-label identification.
- **Built an internal AI-powered HR service desk** with FastAPI, hybrid BM25/vector retrieval, role-based access, configurable approvals, internal mail and human-reviewed local-LLM drafting.
- **Productionised field systems** through REST integrations, SQLite WAL queues, evidence outboxes, Docker/systemd services, restart-safe supervisors and Linux resource hardening.
- **Evaluated models, runtimes and hardware** across YOLO/YOLOE, InsightFace, OCR, ONNX/TensorRT, Jetson and multi-GPU servers; documented architectures, deployment procedures and troubleshooting guidance.
- **Built open-world security prototypes where labelled data was unavailable** using YOLOE/YOLO-World concepts, prompt variants and local vision-language verification for helmet, mask, weapon and stick scenarios.
- **Led client-facing technical delivery** through on-site calibration, PoC demonstrations, deployment troubleshooting, solution design and technical validation for pre-sales and field teams.

Research Assistant

4IR Research Cell, Daffodil International University

Computational healthcare, spectrogram learning and trustworthy AI

Jun 2024 - Feb 2025
Dhaka

- **Published first-author research in the Journal of Voice** on Bengali voice-based psychological stability assessment using a hybrid ViT-CNN; achieved 91% accuracy and ROC-AUC 0.97.
- **Built an ethically collected Bengali voice research dataset** with 85 participants from two mental-health institutions and supported its release for reproducible research.
- **Strengthened evaluation rigour in follow-up work** through subject-independent partitioning and augmentation-to-raw fine-tuning; DenseNet121 reached 94% accuracy and AUC 0.99.
- **Investigated advanced research pipelines** including DenseNet-ViT-GRU hybrids, noise-robust audio learning, LIME/SHAP/Grad-CAM and medical-image preprocessing.
- **Executed reproducible experimentation workflows** with GPU/TPU training, cross-validation, multi-model comparisons, metric analysis and research documentation.

PROJECTS

- **Brac CIN AI Surveillance System - Brac Bank PLC.** Multi-module surveillance architecture for a 1,000-camera rollout; the unified registry snapshot covers 383 cameras and integrates shared RTSP decoding, GPU-aware scheduling, health monitoring, biometric identity, behavioural analytics, suspicious-object detection and Linux resource hardening.
- **Track My Container - KDS Logistics Ltd..** Jetson Orin Nano pipeline combining YOLO-OBb, PaddleOCR, horizontal pair reconstruction, vertical-character rearrangement, ISO 6346 check-digit validation, prefix correction, thermal controls and ESP32/UART telemetry for operational container identification.
- **Argus Automata - Unilever Bangladesh Ltd..** Industrial safety analytics for overspeed, human-vehicle and vehicle-vehicle collision risk, PPE, work-at-height compliance, throwing and intrusion using homography, custom tracking, Kalman/EMA filtering, time-to-collision rules, persistence and evidence-backed alerts.
- **AI Leaf SOP Checker - British American Tobacco Bangladesh.** Four-line, paired-camera bale counting and SOP monitoring for bale opening, layer-by-layer and moisture checks, plus weighing-stage hand activity; used segmentation/OBB models, line-crossing logic, camera-seam merging, persistent counters and retryable evidence delivery.
- **Bangla ANPR & Smart Parking - Enterprise Solution.** Track-centric YOLO-OBb and PaddleOCR with vehicle association, ROI-gated crop sessions, Bengali district/category validation, multi-frame confidence voting and cross-gate deduplication; designed the complete allocation, occupancy, parking-session, fare and barrier workflow.
- **Showroom AI - Runner Automobiles Ltd..** Retail-intelligence suite with employee/customer recognition, self-growing repeat-customer face galleries, custom motorcycle-model detection, visitor and product entry/exit, footfall, service-delay, customer-product interaction and shutter-state analytics.
- **Mobil Product Inspector - Mobil Bangladesh Ltd..** Operator-assisted product identification using three YOLO-OBb models, orientation-corrected crops, multi-image quality ranking, RapidOCR/barcode processing, 22-path manufacturing-code enhancement and consensus, product-catalog matching and SQLite evidence records.
- **Staircase Safety Monitoring - Unilever Bangladesh Ltd..** Pose-based industrial compliance system combining YOLO person detection, MediaPipe landmarks, calibrated handrail polygons and wrist-to-ear phone-use heuristics with scene-level temporal stabilization, explainable overlays, RTSP recovery and evidence/API delivery.
- **Bondstein HR Assist - Internal HR Team.** FastAPI and SQLite/WAL HR service desk with OCR-indexed policies, FTS5 BM25/vector retrieval fused by RRF, Bengali/Banglish support, five-role RBAC, segregation of duties, dynamic requisitions, configurable approvals, internal mail, analytics and human-reviewed AI drafting.
- **CV Agent Ecosystem - Personal Project.** LangGraph-based 13-agent engineering workflow that converts a CV product brief into product requirements, architecture, data strategy, MLOps, code, tests, documentation and operational assets, with typed shared state and human approval gates for high-risk actions.
- **VisionGuard - Personal Project.** Bun/TypeScript and SolidJS multi-camera control plane with H.264/WebSocket streaming, ROI and task administration, persistent Python ML workers, local/remote inference, model lifecycle management, detection evidence, face profiles, VLM agents, analytics and notifications.
- **Unified AI OS - Personal Project.** FastAPI/React research cockpit coordinating Claude, Gemini, ChatGPT and Perplexity through persistent Playwright sessions, with local Ollama tool routing and synthesis, parallel execution, provider-specific prompting, SSE progress, raw response views and SQLite conversation storage.
- **Body-Cam Crowd & Weapon Analytics - Personal Project.** Custom four-class YOLO, DeepSORT and Depth Anything V2 pipeline for depth-aware crowd clustering, graph-based grouping, temporally smoothed density heatmaps and weapon-associated suspect escalation in body-camera, recorded-video or RTSP footage.

PUBLICATIONS

- **Mental Health Diagnosis From Voice Data Using Convolutional Neural Networks and Vision Transformers - Journal of Voice, Elsevier; accepted 2024.** First author. Hybrid ViT-CNN on Bengali log-mel spectrograms; 91% accuracy and ROC-AUC 0.97. DOI: 10.1016/j.jvoice.2024.10.010
- **A Novel Transfer Learning Approach for Mental Stability Classification from Voice Signal - arXiv preprint.** First author. Subject-independent evaluation; DenseNet121 achieved 94% accuracy and AUC 0.99 through augmentation pretraining followed by raw-domain fine-tuning.
- **Using Computer Vision for Skin Disease Diagnosis in Bangladesh - arXiv preprint.** First author. Compact DCNN benchmarked against AlexNet, ResNet, VGG16, DenseNet and MobileNet on HAM10000, emphasizing computational efficiency.

CERTIFICATIONS

- IBM Data Science: What is Data Science? and Tools for Data Science (Coursera)
- Generative AI Productivity Skills - Microsoft and LinkedIn | Brain-Computer Interface (BCI) - Pantech.AI Academy

LEADERSHIP & SERVICE

Executive Member, Computer Programming Club, DIU | Red Crescent Youth Volunteer, Bangladesh Red Crescent Society (2021-2022) | Vice President, Science Club, BAF Shaheen College Dhaka (2017-2018)

LANGUAGES

Bangla - Native | English - Fluent; IELTS Academic 6.5 (L 6.5, R 7.0, W 6.0, S 6.5)

REFERENCE

Md. Ashfaquul Haque - Senior Software Engineer (AI & ML), Bondstein Technologies Ltd. | ahq.john@gmail.com | +880 1792895969